

Xiang Ge (Xiangge) Luo

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Research interests

Probabilistic graphical models, stochastic processes, population dynamics, phylogenetics, evolutionary theory

Education

ETH Zurich

PHD IN COMPUTATIONAL BIOLOGY - ETH MEDAL FOR OUTSTANDING THESIS

- Thesis: Modeling tumor mutation trees for evolution-guided precision oncology
- Advisor: Prof. Dr. Niko Beerenwinkel | Second advisor: Dr. Jack Kuipers
- Doctoral committee: Prof. Dr. Ben Raphael, Prof. Dr. Jasmine Foo

Basel, Switzerland

2020.11 - 2025.12

ETH Zurich

MASTER OF SCIENCE IN STATISTICS

- GPA: 5.88/6 – Graduated with distinction
- Thesis: Learning Bayesian networks from ordinal data
- Advisor: Dr. Jack Kuipers

Zurich, Switzerland

2018.09 - 2020.09

University of Waterloo

BACHELOR OF MATHEMATICS

- GPA: 92% – Graduated with distinction, Dean's Honours List

Waterloo, Canada

2013.09 - 2018.04

Research Experience

Computational Biology Group, ETH Zurich

POSTDOCTORAL RESEARCHER

- Advisor: Prof. Dr. Niko Beerenwinkel
- Developing an evolutionary forecasting framework for treatment resistance based on multitype branching processes, integrating longitudinal single-cell phylogenies from 231 acute myeloid leukemia patients across diagnosis, remission, and relapse.
- Designing an interactive web tool to visualize patient-specific fitness landscapes and forecast survival outcomes under alternative therapeutic strategies.

Basel, Switzerland

2026.03 - present

Computational Biology Group, ETH Zurich

PHD RESEARCHER

- Advisor: Prof. Dr. Niko Beerenwinkel
- Developed mathematical and statistical models to infer cancer evolution from cross-sectional single-cell phylogenies, aiming to predict drug resistance and facilitate evolution-guided precision oncology.

Basel, Switzerland

2020.11 - 2026.02

Institute of Molecular Systems Biology, ETH Zurich

RESEARCH ASSISTANT

- Advisor: Prof. Dr. Mattia Zampieri
- Analyzed E. coli drug response metabolomics data to investigate regulatory and functional relationships between metabolites.

Zurich, Switzerland

2019.03 - 2019.07

Seminar for Statistics, ETH Zurich

STATISTICS LAB RESEARCHER

- Advisor: Dr. Markus Kalisch
- Analyzed mouse experimental data using mixed models to assess bone loss and identify early osteoporosis markers.

Zurich, Switzerland

2019.03 - 2019.06

Department of Statistics and Actuarial Science, University of Waterloo

UNDERGRADUATE RESEARCHER

- Advisor: Prof. Dr. Alexander Schied
- Derived the Nash equilibrium for risk-averse investors in a market impact game with transient price impact and quadratic transaction costs over finite and infinite horizons.

Waterloo, Canada

2017.09 - 2017.12

Publications

Xiang Ge Luo, Jack Kuipers and Niko Beerenwinkel. Numerical approximations of population size distributions for multi-type branching processes. *Manuscript in preparation*, 2026.

Xiang Ge Luo, Jack Kuipers, Kevin Rupp, Koichi Takahashi and Niko Beerenwinkel. Bayesian inference of fitness landscapes via tree-structured branching processes. *Bioinformatics*, 41:i160-i169, 2025. (**ISMB/ECCB 2025**, **Best Paper Award**, acceptance rate 17%)

Xiang Ge Luo, Jack Kuipers and Niko Beerenwinkel. Joint inference of exclusivity patterns and recurrent trajectories from tumor mutation trees. *Nature Communications*, 14:3676, 2023. (**RECOMB 2022**, acceptance rate 21.3%; **SIB Remarkable Outputs 2023 Award**)

Xiang Ge Luo, Giusi Moffa and Jack Kuipers. Learning Bayesian networks from ordinal data. *Journal of Machine Learning Research*, 22:1-44, 2021.

Xiangge Luo and Alexander Schied. Nash equilibrium for risk-averse investors in a market impact game with transient price impact. *Market Microstructure and Liquidity*, 5:2050001, 2020.

Awards & Scholarships

2026	ETH Medal for outstanding doctoral thesis , ETH Zurich	Switzerland
2025	ISMB/ECCB 2025 Ian Lawson Van Toch Memorial Award for Best Student Paper , ISCB	United Kingdom
2025	ISMB/ECCB 2025 Conference Fellowship & Dependent Care Grant , ISCB (USD 950)	United Kingdom
2024	SIB Remarkable Outputs 2023 Award , Swiss Institute of Bioinformatics	Switzerland
2018	Undergraduate Student Research Award , NSERC (CAD 4,500)	Canada
2018	President's Research Award , University of Waterloo (CAD 1,500)	Canada
2016	President's International Experience Award , University of Waterloo (CAD 1,500)	Canada
2014	Isabel Farrar Undergraduate Entrance Scholarship , University of Waterloo (CAD 4,000)	Canada
2014	President's Scholarship of Distinction , University of Waterloo (CAD 2,000)	Canada

Conferences, Seminars & Workshops

2026.06	Poster , Ascona Workshop 2026 • Numerical approximations of population size distributions for multi-type branching processes	Ascona, Switzerland
2025.07	Contributed talk , ISMB/ECCB 2025 Conference • Bayesian inference of fitness landscapes via tree-structured branching processes	Liverpool, UK
2025.04	Poster , 6th Zurich Precision Oncology Symposium • FiTree: fitness inference from single-cell phylogenies applied to acute myeloid leukemia data	Zurich, Switzerland
2025.03	Invited talk , D-BSSE Departmental Seminar Series, ETH Zurich • Fitness inference from single-cell phylogenies using tree-structured branching processes	Basel, Switzerland
2025.02	Contributed talk , PhyloBasel Seminar Series, ETH Zurich • Bayesian inference of fitness landscapes via tree-structured branching processes	Basel, Switzerland
2024.06	Award talk , SIB Days – The Swiss Bioinformatics Summit • Mining tumor mutation trees with TreeMHN for evolution-guided precision oncology	Virtual
2023.06	Contributed talk , SKINTEGRITY.CH Annual Retreat • Joint inference of exclusivity patterns and recurrent trajectories from tumor mutation trees	Spiez, Switzerland
2022.06	Contributed talk , Mutual Hazard Networks Workshop • Joint inference of exclusivity patterns and recurrent trajectories from tumor mutation trees	Virtual
2022.05	Contributed talk , RECOMB 2022 Conference • Joint inference of exclusivity patterns and recurrent trajectories from tumor mutation trees	San Diego, USA
2022.04	Contributed talk , OLISSIPO Exchange Week • Joint inference of exclusivity patterns and recurrent trajectories from tumor mutation trees	Lisbon, Portugal
2022.03	Poster , Ascona Workshop 2022 • Joint inference of exclusivity patterns and recurrent trajectories from tumor mutation trees	Ascona, Switzerland

Academic Services

Journal Reviewer for Genome Research, Bioinformatics

Conference Reviewer for RECOMB, ISMB

Teaching

Spring 2023	Teaching Assistant , Statistical Models in Computational Biology	<i>ETH Zurich</i>
Spring 2022	Head Teaching Assistant , Statistical Models in Computational Biology	<i>ETH Zurich</i>
Spring 2022	Teaching Assistant , LSZGS Advanced Block Course Computational Biology	<i>ETH Zurich</i>
Spring 2021	Teaching Assistant , Statistical Models in Computational Biology	<i>ETH Zurich</i>

Supervision

2025	Marie Dogo , Master Student, Mines Paris - PSL
2023	Laura Quintas , Master Student, INESC-ID (Co-supervised with Monica Baciu-Dragan)
2023	Laurenz Keller , Bachelor Student, ETH Zurich (Co-supervised with Pawel Piotr Czyz)
2022	Jiayi Wang , Master Student, ETH Zurich (Co-supervised with Pawel Piotr Czyz)

Languages and Skills

Coding	Python, R, C++, MATLAB, Git, Snakemake, bash, $\text{\LaTeX}$
Languages	English (working proficiency), Mandarin (native), Cantonese (native)